



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 2

Prepared by: Mohamed B Sirajuddeen

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Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats, researchers, and entrepreneurs in the field of computer science and engineering.

Mission

- To impart quality education in Computer Science and engineering by means of learning techniques and value added courses.
- To inculcate professional excellence and research culture by encourage projects in cutting-edge technologies, through industry interaction.
- To build leadership skills, ethical values and team work among students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

1. ABAP/4 Workbench Tools

→ ABAP workbench is a graphical programming environment in the SAP R/3 system to develop different applications by using the ABAP language.

→ It provides different tools, such as ABAP dictionary, ABAP editor and screen painter, to create ABAP application.

→ The ABAP workbench tools are used to write ABAP code, design the screen and create user interfaces. The following are different tools provided by ABAP workbench.

- ABAP dictionary.
- ABAP ~~Builder~~ Editor
- Class builder
- Function Builder
- Screen painter
- Menu painter
- Object Navigator
- Maintain Message
- ABAP Test element
- Maintain Transaction

ABAP dictionary:

→ ABAP dictionary is one of the important tools of the ABAP workbench.

→ It is used to create and manage data definitions without redundancies. ABAP dictionary always provides the updated information of an object to all the system components.

→ The presence and role of ABAP dictionary ensures that data stored in a SAP system is consistent and secure.

→ The data definitions stored in ABAP dictionary allow you to create various objects, such as tables and views.

The objects stored in ABAP dictionary are logically related to each other in the context of an application.

The initial screen of ABAP dictionary provides the following object types:

- Database table: Creates table definitions in ABAP dictionary, independent of any database.

The created table definition can then be used to create a table in the database.

- View: Creates view definition in ABAP dictionary. A view is a virtual table that does not store the data physically.

- Data type: Creates a definition of a user defined type in ABAP dictionary. The created definition of a user defined type can be used in a program to define data types and data objects.

- Type group: Creates group of data types in ABAP dictionary.

- Domain: Creates domains in ABAP dictionary.

A domain is used to describe the technical attributes of a field, such as range of values.

- Search help: Creates a help document on called Input help for fields.
- Lock object: Creates a local or local objects that helps synchronize the access of multiple users simultaneously.

2. ABAP Editor

ABAP Editor is used to create ABAP program by writing code. The ABAP editor to define the class methods, function modules, screen flow logic, type groups and logical databases for the ABAP program.

- Front-end editor (New)

- Front-end editor (old)

- Back-end editor.

Front-end Editor (New):

- Front-end editor (New) is the latest programming tool to create application in an SAP systems.
- It provides various new features, such as code hints, syntax highlighting, and auto-completion of language structures and elements.

Front-end editor (old):

- As stated earlier, Front-end editor (New) is used to edit the source code of the development objects.
- The classic mode of Front-end editor (old) is also used to edit the development objects, such as ABAP programs, function modules, and type groups.

Back-end editor:

- Back-end editor, another classic editor, allows user to edit ABAP code.

→ This editor is line-based and is used to perform normal editor functions, such as cut, copy and paste.

3. Class Builder

→ Class Builder is a tool of ABAP workbench used to create, define, change and test the global ABAP classes and interfaces.

→ ABAP classes and interfaces are object types defined in repository browser.

The following actions by using the class Builder tool:

- Displaying an overview of global object types and their relationship in class browser.
- Maintaining the already existing global classes or interface.
- Creating new global classes and interfaces.
- Implementing inheritance between two or more global classes.
- Creating compound interfaces.
- Creating and specifying the attributes, methods and events.

4. Function Builder

→ Function Builder plays an important role in defining and maintaining the ABAP functional modules.

→ They are stored in function groups, which act as containers for functions modules that are logically related to each other.

→ The function Builder tool is used to create a function group and function module.

5. Screen painter

→ Screen painter is used to design and manage screen and their elements. It facilitates users to create GUI screens for transactions.

Layout editor: Manages a screen's layout. The screen layout describes both the screen elements and their layout.

Attributes: Manage screen's attributes. Screen attribute determine the type of a screen and the program to which the screen belongs.

6. Menu painter

→ Menu painter is used to design user interfaces for ABAP programs.

→ The main components of Menu painter are GUI status, menu bars, menu lists, F-key settings, functions and titles.

→ In other words, using Menu painter, you can create custom menus in SAP.

7. Object Navigator

→ Object Navigator is another important tool of ABAP workbench.

→ It helps create objects in an SAP system, and navigate from one object to another.

→ It acts as the central area of ABAP workbench, where you can access any object of an SAP system.

- Select a browser and navigate in the object list
- Use the tool for development objects.
- Navigate from one window to another window
- Perform syntax checks in the integrated window
- Open an object in a new session using an additional dialog box.

8. Message Maintenance

→ Messages help an SAP system communicate with the users.

→ For eg: error messages display if the users make invalid entries on a screen, making the users aware that a wrong entry has been made.

→ Each message class has an ID and usually contains a set of messages.

9. ABAP Text element:

→ The text elements of program texts in different languages, can be maintained easily with the help of a tool known as text element.

→ Creation, maintenance and translation of the text elements can be done with the help of the text

- Text symbols
- List and column header.
- Selection texts.

10. Maintain transaction

- Maintain transaction is an ABAP workbench tool used to create and manage user defined transactions.
- Transaction screen contains a transaction code. transaction in the transaction code field.

2) ABAP Syntax and datatypes

- ABAP programs are made up of individual statements.
- The first word in the statement is called an ABAP keyword.
- Each statement must end with a period.
- Statements can be indented.
- Can have multiple statements in a single line.
- Every statement provided in the ABAP programming language has a predefined syntax associated with it.

Data types:

→ Data types describe data objects ABAP provides predefined data types that can be used in an ABAP program.

→ The following are the three kinds of data types.

1. Elementary types
2. Complex types
3. Reference types.

	Term	Description
1.	Statement	Specific that ABAP statements and classes are given in uppercase, for is not mandatory to use the ABAP keywords in uppercase.
2.	Operand	Specifies that the operands used in the statements are in lowercase.
3.	[]	Specifies that the element enclosed in the square bracket is optional.
4.		Specifies that there are elements on both sides of this symbol.
5.	()	Specifies that parenthesis is a part of the syntax of a statement and must be used while using the statement.
6.	,	Separate one or more variable that you need to specify in a statement.
7.	f1, f2	Represents variables indicated with indices list as many variables in a program you want.
8.		Represents the rest of the code is a program besides the syntax of ABAP statements.

Elementary types

→ Elementary types are the smallest undirected unit of predefined data types. These elementary types are already defined in the SAP system and are visible in all ABAP program.

→ ABAP programs use these predefined elementary types to define local data types and data objects used in a program.

→ The following are two types of predefined elementary data types:

- * Fixed-length

- * Variable-length.

→ These types are not derived from other types. The main difference between the fixed and variable length types is that the memory space required by the data objects of variable-length-data types.

The fixed length data types

→ As the name suggests, the length of the fixed length data types are fixed at runtime.

→ The fixed length data types are:

1. Character types
2. Hexadecimal types
3. Numeric types.

Variable - length data type

→ A variable - length data type is used for data objects whose length cannot be fixed as the length varies according to the specified data.

→ The variable length data type is of two types - STRING and XSTRING, described in below table.

Fixed Length data Types:

Fixed - Data types	Initial Field Length (in bytes)	Valid Field Length (in bytes)	Initial Values	Description.
Numeric types.				
I	4	4	0	Specifies an integer (whole number)
F	8	8	0	Represents a floating point number
P	8	1-16	0	Specifies a packed number
Character type.				
C	1	1-65535	'.....'	denotes a text field (alphanumeric characters)
D	8	8	'00000000'	Specifies a date field.

Variable-length data type:

Variable length data type	Description
STRING	Specifies a sequence of character with variable length. A string can contain any number of alphanumeric character. The length of the string can be calculated by multiplying the number of character with the length required for the internal representation of a single-character.
X STRING	Represents a string used for byte string of a hexadecimal type. A byte string has a variable string and can contain any number of bytes.

3. Architecture of SAP R/3

→ The SAP R/3 system is based on the three tier architecture of the client-server model.

→ SAP R/3 architecture basic system forms a central platform within the R/3 system.

→ The link between these systems is established with the help of a network.

→ The SAP R/3 system is implemented in such a way ~~in individual~~ that the presentation, Application and database layers are individual computers in the SAP R/3 architecture.

→ The SAP R/3 system consists of the following three types of views.

→ Logical view.

⇒ Software-oriented view

→ User-oriented view

The Logical view

→ The logical view represents the functionality of the SAP system. In this context, the R/3 basis component controls the functionality and proper functioning of the SAP system.

→ Therefore, in the logical view of the SAP R/3 system, the service provided by the R/3 SAP application.

Description of Various components

→ Kernel and basis services

→ ABAP workbench services

→ Presentation component services.

The Software-oriented view

→ The software-oriented view displays various types of software components that collectively constitute the SAP R/3 systems.

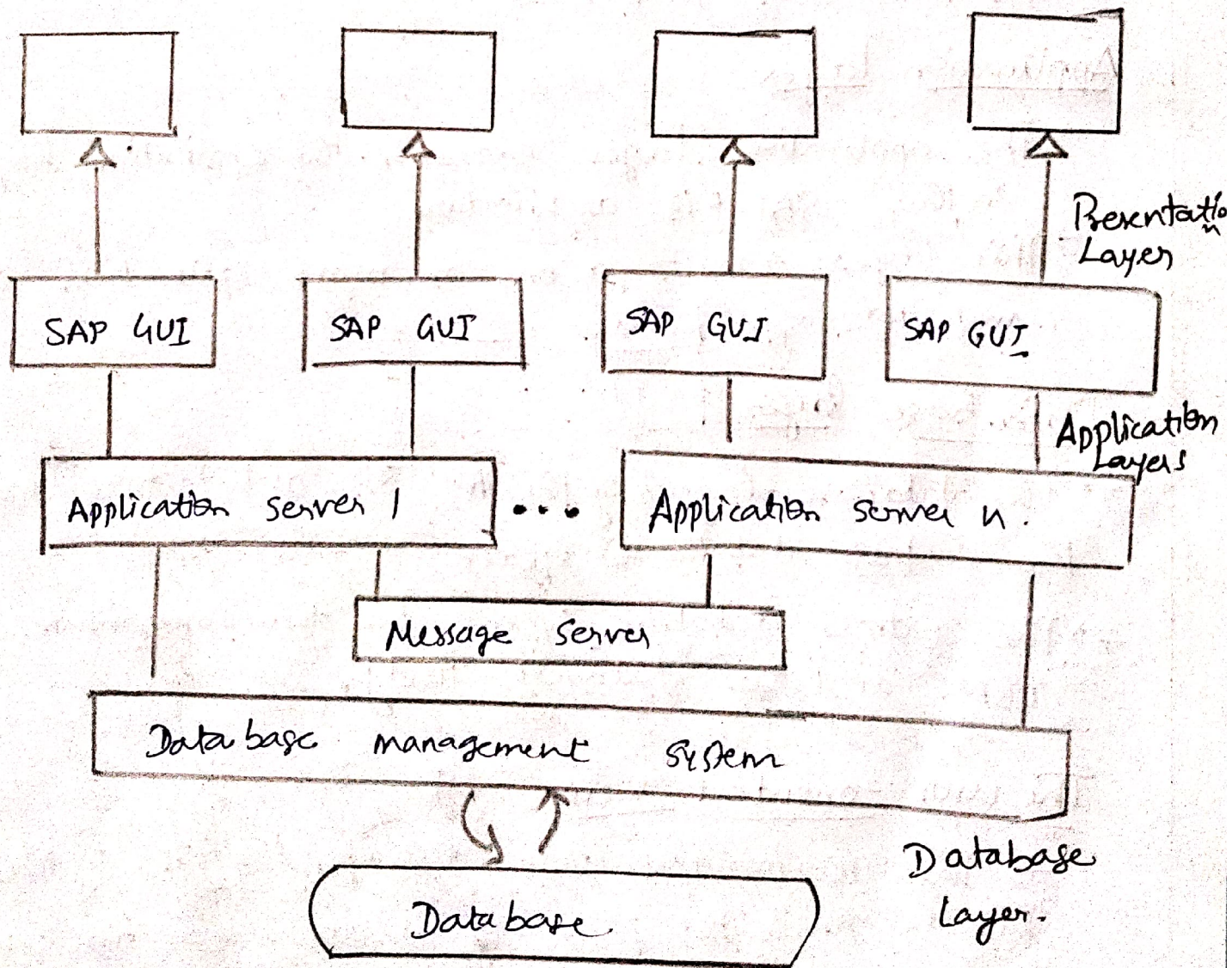
→ It consists of SAP graphical user Interface (GUI) components and application servers, as well as a message server, which make up the SAP R/3 system.

→ Since the SAP R/3 system is a multi-tier client-server system, the individual software components are arranged in tiers.

→ These components act as either clients or servers, based on their position and role in a network.

→ Software-oriented view of the SAP R/3 systems consist of the following three-layers.

- Presentation-layer
- Application-layer
- Data base-layer.



Presentation Layer

- The presentation layer consists of one or more servers that acts as an interface between the SAP R/3 system and its users, who interact with the system with the help of well defined SAP GUI components.
- For eg: Using these components users can enter a request to display the contents of a database table
- The presentation layer then passes the request to the application server.

Application layer

- The application layer executes the application logic in the SAP R/3 architecture.
- This layer consists of one or more application servers and message servers.

Database layer

- The database layer of the SAP R/3 architecture comprises the central database system.
- The central database system has two components, DBMS and the database itself.

The user -oriented view

- The user oriented view displays the GUI of the R/3 system in the form of windows on the screen.
- These windows are created by the presentation layer.
- To view the windows the user has to start the SAP GUI utility, called the SAP login program.